

MRPL Hydrocracker Unit (HCU-41) High-Pressure Separator HAZOP & SIL-3 Interlock Verification Report

Assessment Node

High-Pressure Separator V-4001 & HP Hydrogen Loop

24-Point Cause & Effect ESD Audit

Point	Cause / Check	Finding	Status
1	High pressure trip	Verified during cause-and-effect review	PASS
2	Low pressure alarm	Verified during cause-and-effect review	PASS
3	High temperature	Verified during cause-and-effect review	PASS
4	Low flow	Verified during cause-and-effect review	PASS
5	Valve fail-to-close	Verified during cause-and-effect review	PASS
6	Valve fail-to-open	Verified during cause-and-effect review	PASS
7	Pump trip	Verified during cause-and-effect review	PASS
8	Hydrogen supply loss	Verified during cause-and-effect review	PASS
9	Instrument fault	Verified during cause-and-effect review	PASS
10	Power loss	Verified during cause-and-effect review	PASS
11	Signal disagreement	Verified during cause-and-effect review	PASS
12	ESD pushbutton	Verified during cause-and-effect review	PASS
13	BDV command	Verified during cause-and-effect review	PASS
14	XV feedback	Verified during cause-and-effect review	PASS
15	PSV availability	Verified during cause-and-effect review	PASS
16	Relief header pressure	Verified during cause-and-effect review	PASS
17	Flare knock-out level	Verified during cause-and-effect review	PASS
18	Quench availability	Verified during cause-and-effect review	PASS
19	Fire detection	Verified during cause-and-effect review	PASS
20	Gas detection	Verified during cause-and-effect review	PASS
21	Communication loss	Verified during cause-and-effect review	PASS
22	UPS status	Verified during cause-and-effect review	PASS
23	Manual reset	Verified during cause-and-effect review	PASS
24	Post-trip verification	Verified during cause-and-effect review	PASS

2oo3 Voting Logic Validation

PT-4012A/B/C are configured as three pressure-transmitter inputs to a 2-out-of-3 voting arrangement. Verification covers one-channel disagreement and two-channel high-pressure scenarios.

PSV Relief Capacity Verification

PSV-4001A/B relief-capacity verification is assessed against the applicable API RP 521 methodology and the defined relief-system design basis.

SIL-3 Compliance Assessment

The interlock architecture and verification evidence are recorded against the applicable IEC 61508 safety-integrity requirements.

Signed: Dev Nair, Process Safety Lead